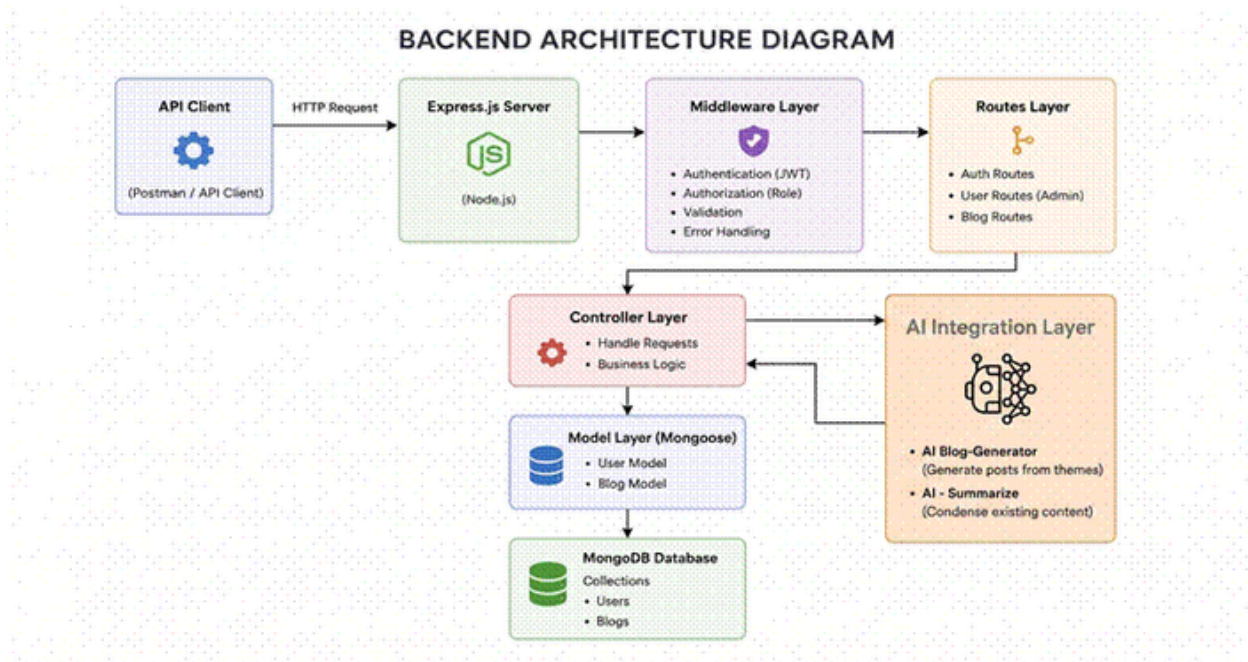


Phase 3
Project Design Phase

Phase 3: Project Design Phase



The BlogNest application exposes structured endpoints handled by a layered server architecture model. Incoming requests undergo filter processing before interacting with data services.

Components Definition

- **Interface:** API tools like Postman query distinct route paths to extract JSON entities.
- **Express Server Gateway:** Monitors network ports, maps explicit HTTP methods (GET, POST, PUT, DELETE), parses JSON payloads, and handles CORS headers.

- **Authentication & Role Matrix Middleware:** Extracts tokens from headers, validates user context signatures, and evaluates if the specific role satisfies authorization thresholds for guarded resource paths.
- **Granular Services (Logic Modules):** Enforces isolated domain rules for blog lifecycles (Draft, Pending Approval, Scheduled, Published), comment queues, and system reporting indicators.
- **Database Interface Layer:** Mongoose translates operational application logic into performant native MongoDB queries.
- **AI Flow (Service Integration):** Provides intelligent content automation, including automated blog post generation and real-time content summarization. This layer connects to external AI models via specialized endpoints (/api/ai/generate-blog and /api/ai/summarize), allowing the controller layer to offload complex text processing tasks and enhance content creation workflows.
- **BlogGenerator with AI:** A specialized service that automates the generation of blog content, which integrates with the AI Flow service to streamline the drafting process.